Light

# GCSE Computer Science - Embedded Systems (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What characteristic makes a computer system 'embedded'?

- ○ **A.** It is designed for a specific dedicated purpose within a larger device
- ○ **B.** It must be connected to the internet
- ○ **C.** It has a large screen display
- ○ **D.** It runs Windows operating system

Q2 (1 mark)

Which device is a general-purpose (non-embedded) computer system?

- ○ **A.** A digital watch
- ○ **B.** A car's airbag controller
- ○ **C.** A microwave timer
- ○ **D.** A desktop PC

Q3 (1 mark)

A modern refrigerator with temperature controls and ice maker contains which type of system?

- ○ **A.** No computer system at all
- ○ **B.** A general-purpose gaming computer
- ○ **C.** An embedded system controlling temperature and ice production
- ○ **D.** A supercomputer for complex calculations

Q4 (1 mark)

Can users install new applications on a typical embedded system like a washing machine controller?

- ○ **A.** Yes, from any app store
- ○ **B.** Yes, but only paid applications
- ○ **C.** No, the software is fixed by the manufacturer for its specific function
- ○ **D.** Yes, using a USB drive

Q5 (1 mark)

Why do most embedded systems not have a keyboard and mouse?

- ○ **A.** Their dedicated functions do not require complex text input or cursor control
- ○ **B.** Keyboards and mice have not been invented yet
- ○ **C.** Embedded systems cannot detect any user input
- ○ **D.** All embedded systems must be voice-controlled

Q6 (1 mark)

A car's fuel injection system that controls precisely how much petrol enters the engine contains which type of system?

- ○ **A.** A laptop computer
- ○ **B.** No computer system
- ○ **C.** A smartphone
- ○ **D.** An embedded system for precise fuel delivery control

Q7 (1 mark)

Why do embedded systems typically generate less heat than powerful desktop computers?

- ○ **A.** Embedded systems are made from special cold materials
- ○ **B.** They use simpler processors doing less work, consuming less power and generating less heat

- ○ **C.** Heat generation is identical for all computers
- ○ **D.** Embedded systems are always kept in refrigerators

Q8 (1 mark)

What can users do with a general-purpose computer that they cannot do with an embedded system?

- ○ **A.** Install and run different applications of their choosing
- ○ **B.** Nothing - both have identical capabilities
- ○ **C.** Connect to electricity
- ○ **D.** Turn the device on and off

Q9 (1 mark)

A digital camera that captures and stores photographs contains which type of system?

- ○ **A.** A general-purpose laptop
- ○ **B.** No computer system
- ○ **C.** A mainframe computer
- ○ **D.** An embedded system dedicated to photography functions

Q10 (1 mark)

Why do embedded systems tend to have fewer software crashes than general-purpose computers?

- ○ **A.** Embedded systems have no software
- ○ **B.** They run simple, dedicated software with fewer components that could conflict or fail
- ○ **C.** Crashes only happen on devices with screens
- ○ **D.** General-purpose computers are designed to crash frequently

Q11 (1 mark)

How does an embedded system's startup compare to a general-purpose computer booting up?

- ○ **A.** Embedded systems take longer to start because they are more complex
- ○ **B.** Both start at exactly the same speed
- ○ **C.** General-purpose computers start faster due to better hardware
- ○ **D.** Embedded systems start almost instantly because they have no complex boot process

Q12 (1 mark)

A hearing aid that amplifies sound for a person with hearing loss contains which type of system?

- ○ **A.** A desktop gaming computer
- ○ **B.** A general-purpose server
- ○ **C.** An embedded system for audio processing

- ○ **D.** No computer system is needed

Q13 (2 marks)

A car's airbag system must deploy the airbag within 30 milliseconds of detecting a collision. Why is this an example of a real-time requirement?

- ○ **A.** Because the system must complete its task within a strict deadline - any delay could result in injury or death
- ○ **B.** Because the airbag displays the current time
- ○ **C.** Because the system only operates in real life, not in simulations
- ○ **D.** Because 30 milliseconds is a very long time for computers

Q14 (2 marks)

Compare a smart speaker (like Amazon Echo) with a laptop computer in terms of embedded versus general-purpose characteristics.

- ○ **A.** Both are identical general-purpose computers
- ○ **B.** Both are identical embedded systems
- ○ **C.** The smart speaker is more embedded (dedicated to voice commands and specific functions) while the laptop is general-purpose (runs user-chosen applications)
- ○ **D.** Neither contains any computer system

Q15 (2 marks)

Why must safety-critical embedded systems undergo extensive certification processes?

- ○ **A.** Certification is purely a marketing requirement
- ○ **B.** Certification only applies to general-purpose computers
- ○ **C.** All software automatically passes certification
- ○ **D.** Failures in safety-critical systems could cause injury or death, so regulators require proof that the system operates correctly under all conditions

Q16 (2 marks)

An automatic garden watering system uses moisture sensors and controls water valves. Explain how this demonstrates embedded system interaction with the physical world.

- ○ **A.** Embedded systems cannot interact with the physical world
- ○ **B.** The system only displays information on a screen
- ○ **C.** Sensors detect soil moisture (input from physical world), the embedded system processes this data and decides when to water, then actuators open valves (output to physical world)
- ○ **D.** The gardener must manually control all watering

Q17 (2 marks)

Why can manufacturers embed computer systems in inexpensive everyday products like toys and appliances?

- ○ **A.** Embedded systems can use minimal, specialised components tailored to specific functions, reducing cost compared to general-purpose hardware
- ○ **B.** Computer chips are free to produce

- ○ **C.** Only expensive products contain embedded systems
- ○ **D.** Embedding computers always increases product cost significantly

Q18 (2 marks)

Traffic lights at a road junction are controlled by which type of system?

- ○ **A.** A person manually changing lights from a control room
- ○ **B.** No control system - lights change randomly
- ○ **C.** An embedded system that manages light timing sequences and responds to sensors detecting vehicles and pedestrians
- ○ **D.** A general-purpose gaming computer

Q19 (2 marks)

How does updating firmware on an embedded system differ from updating an app on a smartphone?

- ○ **A.** Both update processes are completely identical
- ○ **B.** Firmware updates happen automatically every hour
- ○ **C.** Smartphone apps can never be updated
- ○ **D.** Firmware updates often require careful procedures, may need physical access or special tools, and carry higher risk since failure could render the device non-functional

Q20 (2 marks)

What does embedded system miniaturisation enable that would otherwise be impossible?

- ○ **A.** Miniaturisation has no practical benefits
- ○ **B.** Computers in wearable devices, implantable medical devices, and tiny sensors that could not accommodate general-purpose hardware
- ○ **C.** Only large industrial machines can contain embedded systems
- ○ **D.** Smaller systems always perform worse than larger ones

Q21 (2 marks)

A blood glucose monitor used by diabetic patients to measure blood sugar levels contains which type of system?

- ○ **A.** An embedded system that processes sensor readings and displays results
- ○ **B.** A general-purpose laptop computer
- ○ **C.** A gaming console
- ○ **D.** No computer system is involved

Q22 (2 marks)

What security challenge is introduced when embedded systems are connected to networks?

- ○ **A.** Network connectivity improves security automatically

- ○ **B.** Connected embedded systems can be targeted by cyber attacks, but often have limited resources for security features like encryption and may be difficult to update
- ○ **C.** Embedded systems are immune to all cyber attacks
- ○ **D.** Security is only a concern for desktop computers

Q23 (2 marks)

Some embedded systems use a Real-Time Operating System (RTOS) rather than a full operating system like Windows. Why?

- ○ **A.** RTOS is more expensive and complex than Windows
- ○ **B.** There is no difference between RTOS and Windows
- ○ **C.** Windows is designed specifically for embedded systems
- ○ **D.** RTOS provides predictable timing behaviour and minimal overhead, essential for time-critical embedded applications where Windows' complexity would be inappropriate

Q24 (3 marks)

A commercial aircraft contains hundreds of embedded systems. Analyse why the flight control system is separate from the entertainment system rather than combined into one powerful computer.

- ○ **A.** Combining them would be simpler and safer
- ○ **B.** Separation provides critical fault isolation - entertainment system problems cannot affect flight controls. Flight controls require real-time guarantees and safety certification that entertainment systems would complicate. Different reliability requirements are addressed independently

- ○ **C.** Entertainment systems are more important than flight controls
- ○ **D.** Aircraft do not contain any computer systems

Q25 (3 marks)

In 2016, the Mirai botnet compromised hundreds of thousands of IoT devices to launch massive cyber attacks. Analyse what this reveals about embedded system security challenges.

- ○ **A.** Mirai exploited embedded systems with default passwords, no update mechanisms, and minimal security features. This demonstrated that network-connected embedded devices require security considerations historically unnecessary for isolated systems, but face resource and update challenges
- ○ **B.** IoT devices are immune to cyber attacks
- ○ **C.** The attack only affected desktop computers
- ○ **D.** Cyber security is unrelated to embedded systems

Q26 (3 marks)

Analyse the specific requirements that make embedded systems essential for cardiac pacemakers implanted in patients.

- ○ **A.** General-purpose computers would be equally suitable for implanted pacemakers
- ○ **B.** Pacemakers require extreme reliability (no crashes for years), miniature size (fitting inside the chest), extremely low power (battery lasting 10+ years), instant response to heart rhythms, and biocompatible sealed enclosure - requirements only achievable with dedicated embedded design
- ○ **C.** Pacemakers do not contain any computer systems

○ **D.** Size and power consumption are not important for implanted devices

Q27 (3 marks)

A factory uses embedded systems to control assembly line robots and also general-purpose computers for production planning. Analyse why both types are used rather than just one.

○ **A.** Using both types is unnecessary and wasteful

○ **B.** Factories do not use any computer systems

○ **C.** General-purpose computers are better for all factory tasks

○ **D.** Robot control needs real-time response, deterministic timing, and high reliability for safety - embedded system strengths. Planning requires flexibility, database access, user interfaces, and adaptability to changing requirements - general-purpose strengths. Each type excels at different aspects of factory operation

Q28 (3 marks)

Early cars had no computer systems; modern cars contain over 100 embedded systems. Analyse what benefits this computerisation provides and what challenges it creates.

○ **A.** Adding computer systems provides no benefits to cars

○ **B.** Benefits include improved fuel efficiency, enhanced safety features (ABS, airbags, stability control), reduced emissions, better diagnostics, and advanced features. Challenges include increased complexity, potential security vulnerabilities, software bugs affecting safety, and higher repair costs requiring specialist tools

○ **C.** Computer systems make cars less safe

○ **D.** All modern cars are completely mechanical with no computers

Q29 (3 marks)

Compare the testing requirements for an embedded system controlling a railway signal with testing a mobile game app.

○ **A.** The railway signal requires exhaustive testing of all states and failure modes, formal safety analysis, regulatory certification, and proof that incorrect signalling cannot occur. The game requires functionality testing and user acceptance but failures cause inconvenience, not danger. Safety-critical embedded systems demand far more rigorous testing

○ **B.** Both require identical testing approaches

○ **C.** Mobile games require more testing than railway systems

○ **D.** Neither type of software requires any testing

Q30 (3 marks)

Some modern devices like smartphones blur the line between embedded and general-purpose systems. Analyse whether the embedded/general-purpose distinction remains useful.

○ **A.** The distinction has become completely meaningless

○ **B.** The distinction remains useful: smartphones are general-purpose despite embedded heritage; safety-critical systems remain distinctly embedded with dedicated functions, certified software, and real-time requirements. The categories describe design priorities and constraints that persist even as technology evolves

○ **C.** All computers are now identical embedded systems

○ **D.** The categories never had any meaningful difference

Total: **55** marks • Time: **30 min**

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